Example 9.11

Calculate $\lim_{x\to -1}(x^2-3)^{10}$.

Solution

$$\lim_{x\to -1}(x^2-3) = 1-3 = -2.$$

$$\text{Therefore, } \lim_{x\to -1}(x^2-3)^{10} = \lim_{x\to -1}(x^2-3)\,(x^2-3)\ldots(x^2-3) \text{ (10 times)}$$

$$= \lim_{x\to -1}(x^2-3)\lim_{x\to -1}(x^2-3)\ldots\lim_{x\to -1}(x^2-3) \text{ (10 times)}$$

$$= \left[\lim_{x\to -1}(x^2-3)\right]^{10} = (-2)^{10} = 2^{10} = 1024.$$

Note that $\lim_{x\to -1}(x^2-3)^{10} = \left[\lim_{x\to -1}(x^2-3)\right]^{10}$.

Theorem 9.3

If $\lim_{x\to x_0} f(x)$ exists then $\lim_{x\to x_0}[f(x)]^n$ exists and $\lim_{x\to x_0}[f(x)]^n = \left[\lim_{x\to x_0} f(x)\right]^n$.

Example 9.12

Calculate $\lim_{x\to -2}(x^3-3x+6)\,(-x^2+15)$.

Solution

$$\lim_{x\to -2}(x^3-3x+6) = (-2)^3-3(-2)+6 = -8+6+6 = 4$$

$$\lim_{x\to -2}(-x^2+15) = -(-2)^2+15 = -4+15 = 11$$

$$\lim_{x\to -2}(x^3-3x+6)\,(-x^2+15) = \lim_{x\to -2}(x^3-3x+6)\lim_{x\to -2}(-x^2+15) = 4\times 11 = 44.$$

Example 9.13

Calculate $\lim_{x\to 3}\dfrac{(x^2-6x+5)}{x^3-8x+7}$.

Solution

$$\lim_{x\to 3}(x^2-6x+5) = 3^2-6\times 3+5 = -4$$

$$\lim_{x\to 3}(x^3-8x+7) = 3^3-8\times 3+7 = 10 \neq 0.$$

$$\text{Therefore, } \lim_{x\to 3}\frac{(x^2-6x+5)}{x^3-8x+7} = \frac{\lim_{x\to 3}(x^2-6x+5)}{\lim_{x\to 3}(x^3-8x+7)} = \frac{-4}{10} = -\frac{2}{5}.$$

Caution

Do not use the limit theorem for the quotient if $\lim_{x\to x_0} g(x) = 0$.

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